

EcoSort AI — Agentic Waste Segregation & Recycling Assistant

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EfficientNet-B0 Model • 96.22% Test Acc
12 Classes • 15,515 Images • Multi-Agent System

1. PROBLEM

- Household waste is often incorrectly segregated.
- People are unsure whether waste is recyclable, reusable, or hazardous.
- Lack of simple, accessible guidance for proper waste disposal.
- Manual identification of different waste categories is inconvenient.

2. EXISTING ALTERNATIVES

- Manual waste identification.
- Searching recycling/disposal info online.
- General-purpose recycling guides & websites.
- Guess-based household waste segregation.

3. SOLUTION

- AI-powered image-based waste classification.
- Automatic recycling recommendations.
- Category-specific disposal & safety guidance.
- Agentic AI assistant for waste management.
- Analytics dashboard for scans & history.

4. KEY METRICS

- 96.22%** test accuracy (EfficientNet-B0).
- 12** distinct waste categories classified.
- 15,515** total dataset images evaluated.
- Average prediction confidence per scan.
- Total number of waste scans processed.
- Recycling & disposal recommendations served.

5. UNIQUE VALUE PROPOSITION

“Snap your waste. Know where it belongs.”

- EcoSort AI combines computer vision and agentic AI to identify household waste and provide personalized recycling, disposal, and sustainability guidance through one simple interface.

HIGH-LEVEL CONCEPT

- “AI-powered waste scanner + recycling guide + disposal assistant in one platform.”

6. UNFAIR ADVANTAGE

- Custom-trained 12-class EfficientNet-B0 classifier (96.22% accuracy).
- Integrated multi-agent architecture (Orchestrator + 4 sub-agents).
- Unified classification, recycling, disposal & conversational assistant.
- Context-aware Eco Assistant leveraging detected waste class.
- Verified offline knowledge-base fallback when LLM is unavailable.

8. CHANNELS

- Streamlit interactive web application.
- Educational institutions & student communities.
- Sustainability & eco awareness campaigns.
- Social media & digital awareness platforms.
- Demonstrations & academic project showcases.

7. CUSTOMER SEGMENTS

- Primary Segments**
- Households & individual residential users.
 - Students & educational institutions.
 - Environment-conscious consumers.
 - Housing societies & residential communities.
 - Waste-management & sustainability initiatives.
- Early Adopters & Focus**
- University campuses & eco-club members adopting proactive segregation.
 - Responsible urban households seeking instant binning clarity.

9. COST STRUCTURE

- Model development, transfer learning & GPU training.
- Dataset curation, preprocessing, stratified splitting & storage.
- Cloud hosting, inference compute & API usage.
- OpenRouter / LLM API infrastructure for conversational agent.
- Web application development, maintenance & monitoring.
- Streamlit platform hosting & deployment operations.

10. REVENUE STREAMS

- Academic Prototype**
- Primary goal: social & environmental impact (free non-commercial prototype).
- Potential Future Revenue Models**
- Institutional subscriptions for schools, universities & residential complexes.
 - Municipal & waste-management organization partnerships.
 - Sustainability compliance analytics & segregation audit reports.
 - Customized enterprise solutions for private housing societies.
 - API & service integration for third-party smart waste bins.